

Phu Vinh Nguyen

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EDUCATION

Northeastern University Certificate in Bioinformatics, concentration in programming Relevant Coursework: Python for Bioinformatics, Genomics in Bioinformatics, Algorithms, R programming	Boston, MA May 2025
Uniwersytet Jagielloński Master's in Molecular Biotechnology	Krakow, PL Jul 2021
University of California, Irvine Bachelor of Science in Pharmaceutical Sciences	Irvine, CA Sept 2019

WORK EXPERIENCE

Lab Automation Engineer | Developer, Sanofi Pasteur Jun 2025 – Present

- Developed a form help request desktop app using React and Electron to standardize researcher form submissions and automate internal ticket creation via SMTP email integration.
- Designed, tested, and deployed methods for liquid handler robots to automate lab workflows.
- Delivered biweekly presentations to stakeholders on automation performance, challenges, and improvements supported by technical data.

Automation Engineer, Foundation Medicine Inc Nov 2024 – Apr 2025

- Built a Python GUI app using CustomTkinter to provide a user-friendly interface for backend automation scripts, enabling R&D customers to perform tasks independently.
- Deployed a React troubleshooting app for Hamilton liquid handling robot devices to display key metadata in real time, enabling teams to accelerate debugging and improve workflow reliability.
- Created barcode scanning systems and ETL workflows to integrate sample data into Sapio LIMS, automating sample tracking and reducing manual entry errors.

Lab Automation Engineer, Sanofi Pasteur Oct 2022 – Nov 2024

- Developed a suite of Python scripts to automate routine tasks, including data parsing, file management, and data analysis.
- Trained end-users across R&D teams on automated robotic systems
- Optimized high-throughput Hamilton liquid handling methods, reducing assay processing times by up to 30% in DNA extraction and fluorescence-based workflows.
- Integrated laboratory automation devices into robotic workflows through TCP/IP and APIs, to enable command execution and data exchange across multiple devices.

Senior Research Associate, Dermibiont Inc. Oct 2021 – Sept 2022

- Built data analysis pipelines for Oxford Nanopore Technology (ONT) using open-source bioinformatics tools, including Guppy basecalling and quality control filtering.
- Collaborated with external bioinformatics teams to troubleshoot and optimize Dockerized QC pipelines.
- Automated DNA extraction on Opentrons robot and visualized results using Python and R.

PROJECTS

[Personal Website](#)

- Developed a React portfolio website to showcase professional experience, technical projects, and programming skills.

SKILLS

Languages: Python, JavaScript, SQL, HTML, CSS, Bash

Frameworks and Libraries: React, Express, Electron, CustomTkinter

Technical Tools: Node, Git, Docker, Jira, LIMS (user level), HPC environments

Database: MySQL, Postgres